

Spatial Population Dynamics: Fertility, Migration, and Policies

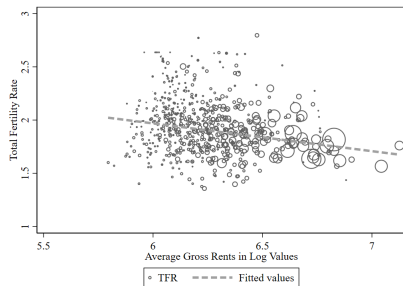
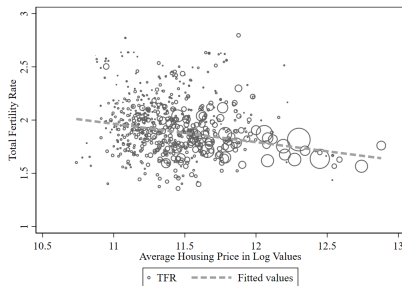
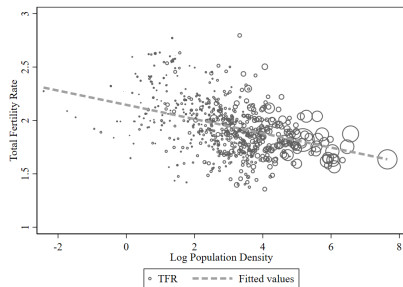
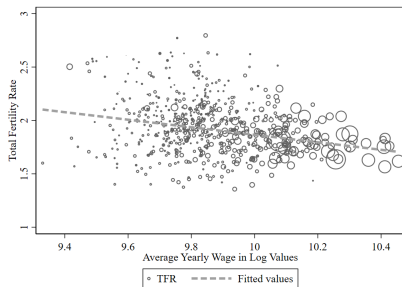
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Motivation

- Fertility has fallen sharply across advanced economies
→ U.S. TFR: 2.1 (1990) → 1.6 (2023)
- Striking empirical regularity: **lowest fertility in most productive cities**
⇒ Spatial reallocation toward productive cities may come at cost of lower aggregate fertility
- Existing explanations (gender roles, labor frictions) are often silent on spatial dimension
- **Key insight:** Fertility should be analyzed as equilibrium outcome of a **spatial household problem**

Motivating Facts



This Paper

1. Develop a [quantitative spatial model](#) integrating endogenous fertility, migration, and housing markets
2. Derive expressions on how local conditions affect fertility and propagate across space
3. Calibrate to [741 U.S. commuting zones](#) and evaluate place-based policies
4. Characterize [optimal interregional transfers](#) under different welfare objectives

Key Findings

1. Housing and children are **strong complements** ($\rho = 0.49$)
 $\implies$ High housing costs reduce both housing consumption and fertility
2. **Housing supply reform** raises aggregate population growth by 1 percentage point ($0.92 \rightarrow 0.93$)
3. Optimal transfers depend on welfare objective:
 - $\rightarrow$ Maximizing **per-capita welfare**: concentrate workers in large CZs
 - $\rightarrow$ Maximizing **total welfare**: retain workers in smaller CZs

Literature

- Quantitative spatial economics (Redding & Rossi-Hansberg 2017, Allen & Arkolakis 2014)
- Economics of fertility (Becker 1960, Barro & Becker 1989)
- Housing and fertility (Dettling & Kearney 2014)
- Housing supply constraints (Hsieh & Moretti 2019)
- Concurrent work: Arkolakis et al. (2026) on South Korea

Roadmap

- Model
- Model mechanisms
- Calibration and model fit
- Counterfactual analysis
- Conclusion

Model

Environment

- N regions, discrete time, two-period OLG
- **Young**: reside with parent, no decisions
- **Adult**: observe location preference shocks, choose:
 - Where to live (migration)
 - Consumption, housing, fertility
- Each region produces differentiated variety (CES aggregation)
- Housing supplied with region-specific elasticity (Saiz 2010)

Equilibrium Definition

Preferences

Adult agent ν moving from o to i obtains tenure status $j \in \{O, R\}$ with probability ζ_i and derives utility:

$$U_{oi,t}^j(\nu) = \frac{B_{i,t}(\nu)}{d_{oi,t}} V_{i,t}^j(\nu)$$

Nested CES utility:

$$V_{i,t}^j = \left[\alpha^{\frac{1}{\gamma}} C^{\frac{\gamma-1}{\gamma}} + (1 - \alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h^{\frac{\rho-1}{\rho}} + (1 - \beta)^{\frac{1}{\rho}} n^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1} \frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}$$

- γ : elasticity between consumption and housing-fertility composite
- ρ : elasticity between housing and children
- $B_{i,t}(\nu)$: idiosyncratic taste shock (Fréchet distributed)

Budget Constraint and Production

Household budget constraint:

$$P_{i,t}C_{i,t}^j + r_{i,t}h_{i,t}^j = w_{i,t}(1 - \chi_{i,t}n_{i,t}^j) + T_{i,t}^j$$

- Children require $\chi_{i,t}$ units of parental time each
- Opportunity cost: $\chi_{i,t}w_{i,t}$ per child

Production:

$$\text{Unit cost: } uc_{i,t} = \frac{w_{i,t}}{A_{i,t}(L_{i,t}^e)^\psi}$$

- $\psi \geq 0$: agglomeration externality
- $L_{i,t}^e = (1 - \chi_{i,t}n_{i,t})L_{i,t}$: effective labor

Housing Supply and Revenue Distribution

Housing supply (Saiz 2010):

$$r_{i,t} = \bar{r}_{i,t} H_{i,t}^{\xi_i}$$

- ξ_i : inverse housing supply elasticity (higher $\Rightarrow$ more inelastic)

Housing revenue distribution:

$$T_{i,t}^O = \frac{1}{\zeta_i} \left[\theta \cdot \frac{r_{i,t} H_{i,t}}{L_{i,t}} + (1 - \theta) \cdot \frac{\sum_{k=1}^N r_{k,t} H_{k,t}}{\sum_{k=1}^N L_{k,t}} \right], \quad T_{i,t}^R = 0$$

- ζ_i : probability of being homeowner in region i
- Local revenue + national redistribution to homeowners

Optimal Fertility

Household optimization yields:

$$n_{i,t} = \frac{1}{\chi_{i,t} w_{i,t}} \cdot \frac{(1 - \beta)(\chi_{i,t} w_{i,t})^{1-\rho}}{\beta r_{i,t}^{1-\rho} + (1 - \beta)(\chi_{i,t} w_{i,t})^{1-\rho}} \cdot \frac{(1 - \alpha) \Psi_i^{\frac{1-\gamma}{1-\rho}}}{\Xi_i} \cdot (w_{i,t} + T_{i,t})$$

Key economic forces:

- Fertility depends on ratio $r_{i,t}/(\chi_{i,t} w_{i,t})$
- When $\rho < 1$: housing and children are complements
- Higher wages: substitution effect $-$, income effect $+$
- Higher rents: price effect $-$, wealth effect $+$

Migration and Population Dynamics

Given Fréchet preferences, migration share from o to i :

$$\frac{L_{oi,t}}{n_{o,t-1}L_{o,t-1}} = \frac{B_{i,t}(d_{oi,t}^{-1}V_{i,t})^\eta}{\sum_{k=1}^N B_{k,t}(d_{ok,t}^{-1}V_{k,t})^\eta}$$

Key features:

- Fertility in numerator through origin's $n_{o,t-1}$
- Fertility in destination's attractiveness $V_{i,t}$
- Rich interaction between demographic and spatial forces
- $\eta > 1$: migration elasticity

Steady State

Population in all regions grows at common rate g :

$$g = \frac{\sum_{i=1}^N n_i L_i}{\sum_{i=1}^N L_i}$$

Implications:

- g is **endogenous**: depends on spatial distribution of fertility
- Policies reallocating population to high-fertility regions raise g
- If $g > 1$: population grows; if $g < 1$: population declines

Model Mechanisms

Local Fertility Elasticities

Proposition: Holding transfers T_i fixed:

$$\varepsilon_{n,w} = \underbrace{-\rho - (\gamma - \rho)\tilde{\omega}_i^n - (1 - \gamma)(1 - \omega_i^C)\tilde{\omega}_i^n}_{\text{substitution effect}} + \underbrace{\frac{w_i}{w_i + T_i}}_{\text{income effect}}$$

$$\varepsilon_{n,r} = \underbrace{\tilde{\omega}_i^h [-(\gamma - \rho) - (1 - \gamma)(1 - \omega_i^C)]}_{\text{price effect}}$$

When transfers respond endogenously to rents:

$$\frac{\partial \log n_i}{\partial \log r_i} = \varepsilon_{n,r} + \underbrace{\frac{T_i}{w_i + T_i} \cdot \varepsilon_{T,r}}_{\text{wealth effect } (>0)}$$

Homeowners vs. Renters

Corollary: The relative fertility of homeowners to renters rises with rents:

$$\frac{\partial \log(n_i^O / n_i^R)}{\partial \log r_i} = \frac{\frac{1}{\zeta_i} T_i}{w_i + \frac{1}{\zeta_i} T_i} \varepsilon_{T,r} > 0$$

Intuition:

- Homeowners receive housing revenue transfers
- Higher rents $\Rightarrow$ larger transfers to owners
- Owners' fertility rises relative to renters'
- Consistent with Dettling & Kearney (2014)

Aggregate Fertility Decomposition

Proposition:

$$\hat{n} = \underbrace{\sum_i \frac{s_i n_i}{\bar{n}} \hat{n}_i}_{\text{intensive margin}} + \underbrace{\frac{1}{\bar{n}} \text{Cov}_s(n_i, \hat{L}_i)}_{\text{extensive margin}}$$

where s_i is population share of region i

Key insight:

- Extensive margin negative when population flows to low-fertility regions
- Pro-fertility policy in expensive cities may reduce aggregate fertility

Fertility as Stabilizing Force

Proposition: Endogenous fertility dampens population responses to local productivity shocks:

$$\hat{L}_j^{SS} \big|_{\text{endogenous } n} < \hat{L}_j^{SS} \big|_{\text{exogenous } n}$$

Mechanism:

- Productivity boom $\Rightarrow$ higher wages, migrants arrive
- Housing costs rise $\Rightarrow$ fertility declines
- Lower fertility partially offsets population growth
- Explains why productive cities haven't grown even faster

Spatial Propagation (1)

Setup: Region j experiences a productivity shock $\hat{A}_j > 0$

Three channels shape GE responses

Trade channel: Price indices respond across trading partners

$$\hat{P}_i = \sum_k \lambda_{ik} \underbrace{(\hat{w}_k - \hat{A}_k)}_{\text{unit cost}} - \psi \underbrace{\sum_k \lambda_{ik} \hat{L}_k^e}_{\text{agglomeration}}$$

Fertility link: Higher fertility today $\Rightarrow$ larger future $\hat{L}_k^e \Rightarrow$ stronger agglomeration $\Rightarrow$ lower prices for trading partners

Spatial Propagation (2)

Migration channel: Population responds to utility and origin demographics

$$\hat{L}_i = \sum_o \pi_{oi} \left[\underbrace{\eta(\hat{V}_i - \bar{\hat{V}}_o)}_{\text{utility-driven}} + \underbrace{\hat{n}_o + \hat{L}_o}_{\text{origin population}} \right]$$

- **Fertility link:** Higher $\hat{n}_o \Rightarrow$ more potential migrants $\Rightarrow$ larger inflows to i

Housing channel: Rents respond to demand and population

$$\hat{r}_i = \xi_i \left(\hat{h}_i + \hat{L}_i \right)$$

- **Fertility link:** Higher fertility $\Rightarrow$ larger $\hat{L}_i \Rightarrow$ higher $\hat{r}_i \Rightarrow$ feedback that *reduces* fertility

Calibration

Data and Parameters

Pre-Determined Parameters	
$\psi = 0.1$	Agglomeration elasticity
$\eta = 1.5$	Migration elasticity (Tombe & Zhu 2019)
$\sigma - 1 = 4$	Trade elasticity
$\theta = 0.5$	Local rental income share
ξ_i	Inverse supply elasticity (Saiz 2010)
Data Sources	
Census 2000	Wages, population, fertility, rents
Commodity Flow Survey	Trade flows (gravity estimation)
Census Migration Tables	County-to-county flows

Preference Parameter Calibration

Identification strategy:

Parameter	Identifying Moment
γ : consumption vs. housing-fertility	$\text{corr}(n_i, w_i)$
ρ : housing vs. fertility	$\text{corr}(n_i, L_i)$
α : consumption share	average fertility
β : housing share	aggregate rent-to-income ratio

Key restriction: Uniform $\chi_i = 0.15$ during calibration

⇒ Forces model to explain variation through wages and rents

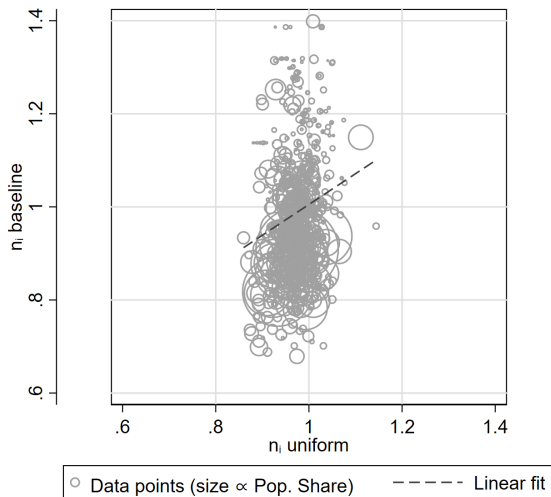
Calibration Results

γ	ρ	α	β
0.91	0.49	0.66	0.60

Key finding: $\rho = 0.49 < 1$

- Strong complementarity between housing and children
- When housing becomes expensive, households cut back on *both*
- Central to understanding fertility-housing relationship

Model Fit: Fertility



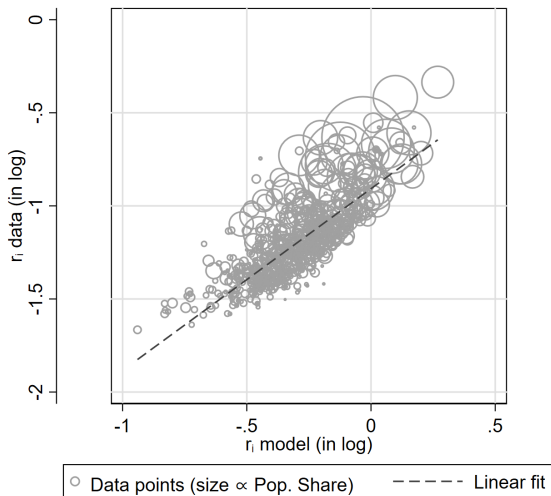
Under uniform χ_i : Slope = 0.66 (s.e. 0.15), $R^2 = 0.03$

Model Fit: Correlations under Uniform χ_i

Dependent Variable: n_i under uniform χ_i			
	(1)	(2)	(3)
$\log(w_i)$	-0.041^{***} (0.008)		
$\log(L_i)$		-0.004^{***} (0.001)	
$\log(r_i)$			-0.154^{***} (0.006)
R^2	0.039	0.038	0.556

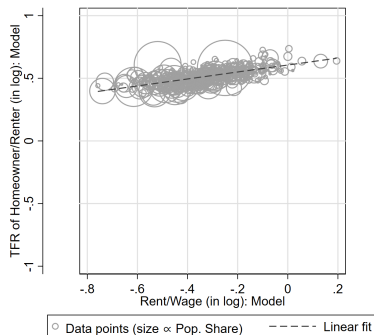
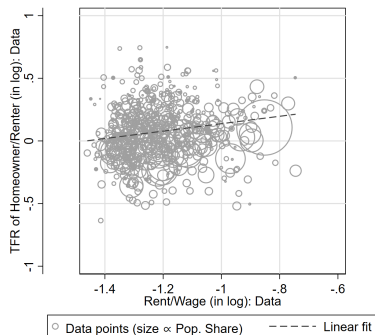
Model reproduces negative correlations from motivating facts

Model Fit: Housing Rents



Slope = 0.98 (s.e. 0.03), $R^2 = 0.61$

Model Fit: Homeowner vs. Renter Fertility



Data: slope = 0.30 (s.e. 0.07); Model: slope = 0.28 (s.e. 0.01)

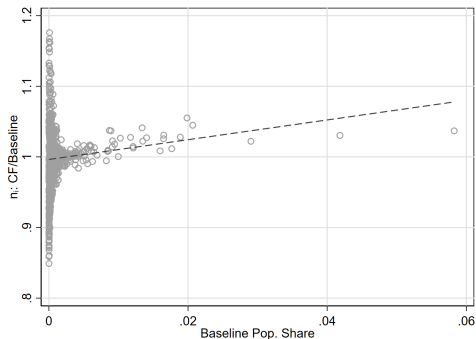
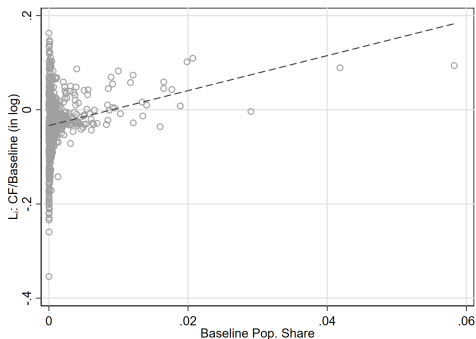
Counterfactual Analysis

Policy Exercise 1: Housing Supply Reform

Experiment: Equalize housing supply regulations to median

Results:

- Population increases in large CZs (migration to now-affordable cities)
- Fertility rises in large CZs (housing-fertility complementarity)



Housing Reform: Aggregate Effects

	g	$std(w_i)$	$std(r_i)$	$std(\log L_i)$
Baseline	0.9188	0.1898	0.1294	1.6328
Counterfactual	0.9288	0.1893	0.1633	1.6537

Results:

- Population growth rate rises by 1 percentage point – modest change because large CZs have low fertility baseline
- Wage dispersion falls (spatial convergence)
- Rent dispersion rises (asymmetric policy effects)

Policy Exercise 2: Optimal Spatial Transfers

Consider transfers $t_n \in [0, 1]$, $\sum_n t_n = 1$:

$$T_n = \theta \cdot \frac{r_n H_n}{L_n} + t_n \cdot \frac{1 - \theta}{L_n} \sum_{k=1}^N r_k H_k$$

Two objectives:

1. Per-capita welfare: $\max W^{ave} \equiv \sum_o \frac{L_o}{\sum_{o'} L_{o'}} \left[\sum_k B_k (d_{ok}^{-1} V_k)^\eta \right]^{1/\eta}$
2. Total welfare: $\max W^{tot} \equiv \sum_o L_o \left[\sum_k B_k (d_{ok}^{-1} V_k)^\eta \right]^{1/\eta}$

Optimal Transfers: Outcomes and Tradeoffs

	W^{ave}	W^{tot}	g	$std(w_i)$	$std(r_i)$
Baseline	81.4	55,434	0.919	0.190	0.129
Opt: Per-capita	98.7	62,422	0.853	0.184	0.193
Opt: Total	96.7	63,290	0.883	0.193	0.201

Key tradeoffs:

- Average utilitarian: concentrate workers in few CZs with high productivity, amenity, and child costs, +21% welfare, but fertility falls ($g = 0.85$)
- Total utilitarian: retain workers across more CZs, +19% per-capita welfare, fertility falls less ($g = 0.88$)
- Optimal design depends critically on welfare objective

Conclusion

Model: Spatial equilibrium with endogenous fertility

- Higher wages reduce fertility (substitution $>$ income effect)
- Higher housing costs reduce fertility (housing-children complementarity)
- Local shocks propagate through trade, migration, and housing channels

Empirics: Housing and children are strong complements ($\rho = 0.49$)

Policy implications:

- **Housing supply reform:** raises both efficiency and fertility
- **Optimal transfers:** per-capita objective concentrates workers but sacrifices fertility; total welfare objective balances growth with population

Extensions

- Heterogeneity in worker skills
- Dynamics of human capital formation
- International migration
- Education competition (cf. Arkolakis et al. 2026)

Appendix

Equilibrium Definition

Given fundamentals $(A_i, B_i, d_{oi}, \tau_{in}, \chi_i, \bar{r}_i)$ and parameters, equilibrium consists of:

1. Wages w_i clear labor market
2. Rents r_i clear housing market
3. Housing supply H_i satisfies inverse supply function
4. Price indices P_i from CES aggregation
5. Fertility rates n_i from household optimization
6. Population L_i from migration decisions